ANDY ZHANG

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QUANT FINANCE EXPERIENCE

AllianceBernstein Holding LP — Nashville, TN
Quantitative Researcher, Associate Officer — Jun 2022 – Present

- Co-developed a long-only systematic European High Yield strategy using the team's proprietary alpha research platform
- Evaluated factor combinations using correlation and information coefficients to determine an optimal combined signal
- Defined risk targets with European HY traders, including OAS, OAD, and OASD, for monthly portfolio rebalancing
- Redesigned the systematic EM debt research framework to accelerate development of new factors and systematic strategies
- Enhanced the fixed income portfolio managers' Python-based key-rate duration (KRD) optimization tool by adding new capabilities including hedging with principal STRIPS and enforcing asset allocation restrictions
- Expanded the batch KRD optimization tool to support concurrent multi-portfolio and intraday hedging, enabling broader adoption across fixed income portfolio management teams
- Constructed a long-only Canadian multi-sector Strategic Portfolio with allocations in US and European fixed income indices
- Developed a constrained mean-variance optimization framework using the team's proprietary Python optimization library

Cleco Corporate Holdings LLC — Cambridge, MA
Quantitative Analyst, Analyst — Nov 2020 – Apr 2021

- Reconciled daily mark-to-market reports for Cleco Cajun, using Allegro Horizon and MarketView
- Produced and automated visualizations including candlesticks, combo charts, and stacked charts using Python's matplotlib and Plotly libraries, as well as VBA
- Supported monthly risk reporting by integrating visualizations into PowerPoint presentations for senior management

Standard Normal Ventures (*start-up energy hedge fund*) — Cambridge, MA
Energy Quant Developer & Trading Analyst, Analyst — Jan 2020 – Aug 2020

- Improved virtual bidding idea generation for ISO-NE by building an HTML-based tool that consolidated feeds on major transmission lines impacted by outages
- Developed data scraping and virtual bidding tools for ISO-NE, NYISO, MISO and ERCOT using Excel with VBA and Python to enhance efficiency, reduce operational risk, and speed up data collection
- Created virtual bidding strategies with senior traders by analyzing fundamental data such as temperature and wind speed and applying statistical analysis using Yes Energy and Python
- Automated tools on an external computer using TeamViewer, VisualCron, and Python's sched module

FINANCIAL ENGINEERING SKILLS

- Implementing ensemble classification models using CatBoost, LightGBM, and XGBoost, with repeated holdout validation, probability blending, and threshold optimization
- Scraping data from dynamic/static websites, wrangling and storing it in databases for use by traders and analysts
- Building Python-based pricing models for asset-backed securities (ABS) and other fixed income and derivative instruments, with a focus on flexibility and computational speed
- Constructing pricing frameworks for vanilla and exotic equity options, as well as interest rate derivatives, using industry-standard models and numerical methods
- Developing DLLs to create efficient, user-friendly tools in Excel VBA for financial analysts and portfolio managers

Technical Skills: MATLAB, Python, R, C++, Bloomberg, Excel VBA, SQL, Git, Azure, AI tools

EDUCATION

Columbia University — New York, NY
M.S., Operations Research — Sep 2017 – Feb 2019

Stevens Institute of Technology — Hoboken, NJ
M.S., Financial Engineering — Aug 2014 – May 2016

- Award: Best Master's Thesis Award

University of Toronto — Toronto, CA
B.Comm., *Specialist in Finance and Economics*, *with distinction* — Sep 2010 – Jun 2014

- Awards: Howard Ferguson Provincial Scholarship; Toronto International Dragon Boat Race Festival 2012 (1st Place in Final E and 2nd Place in Final B)

PUBLICATIONS & INTERESTS

Publications: *"Risk-Averse Dynamic Arbitrage in Illiquid Markets"*, with S. Moazeni and R.A. Collado, *Journal of Risk*, 2018
Interests: Blitz Chess, Puzzles, Popping & Tutting, Traveling, NBA, Dragon Boat Racing, Snowboarding